

Spring data JPA Assignment

Part 1: Entity Creation and CRUD Operations

1. Create Entities:

- a. Create two entities, Book and Author.
 - i. The Book entity should have fields like id, title, genre, and authorId (which should be a foreign key to Author).
 - ii. The Author entity should have fields like id and name.
 - iii. Ensure there is a @OneToMany relationship between Author and Book.

2. Implement CRUD Operations:

- a. Create a Spring Data JPA repository for both entities.
 - b. Implement the following CRUD operations:
 - i. Create a new book or author.
 - ii. Read a list of all books and authors.
 - iii. Update the title of a book and the name of an author.
 - iv. Delete a book or author by id.
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Part 2: Custom Queries with @Query

3. Custom Query for Books:

- a. Write a custom query to find all books written by a specific author using the author's name.
- b. Write a custom query to search for books whose title contains a given string.

4. Custom Query for Authors:

- a. Write a custom query to find authors whose name starts with a specific letter (e.g., "A").
 - b. Write a custom query to return all authors who have written more than a certain number of books (e.g., more than 5 books).
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Part 3: @Transactional and Database Transactions

5. Transactional Method:

- a. Implement a `BookService` class that contains a method for creating a new book and assigning it to an existing author. This method should be transactional, meaning it should either commit both operations (creating a book and linking it to an author) or rollback if any operation fails.
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Part 4: Advanced Queries and Pagination

6. Find Books by Genre:

- a. Create a method in `BookRepository` that retrieves all books of a specific genre. Use pagination to limit the number of books retrieved (e.g., 10 books per page).
- b. Implement this method and expose an API endpoint that returns the paginated list of books by genre.

7. Sorting Query:

- a. Write a query that retrieves a list of books sorted by title in ascending or descending order.
 - b. Implement this query using Spring Data JPA repository methods (e.g., `findAll(Sort.by(...))`).
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Part 5: Entity Relationships

8. Bidirectional Relationship:

- a. Create a `Publisher` entity with fields such as `id`, `name`, and `location`.
- b. Implement a bidirectional `ManyToMany` relationship between `Book` and `Publisher`, where a book can be published by multiple publishers, and a publisher can publish multiple books.
- c. Ensure both sides of the relationship are properly mapped with the `@ManyToMany` and `@JoinTable` annotations.

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spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQLDialect
spring.jpa.hibernate.ddl-auto=validate
spring.jpa.show-sql=true
```